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Classifying Lake Trout vs Smallmouth Bass from Wideband Hydroacoustics (45–170 kHz)

**ETC5543 — Business Analytics Creative Activity
(Single-student project)**

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1 Abstract

Wideband hydroacoustics allows non-invasive monitoring of fish populations, but reliable species-level identification is still challenging without visual confirmation. This project investigates whether frequency-only acoustic signals (45--170 kHz) can accurately classify Lake Trout (LT) and Smallmouth Bass (SMB). Building on earlier work that used a recurrent neural network (RNN) for the same dataset, first replicate that baseline and then extend the analysis using a broader, leakage-safe machine-learning framework.

Summarise each fish's frequency response curve (FRC) using quantiles and the `tsfeatures` time-series descriptors, then apply H2O AutoML across multiple model families with grouped validation by fish identifier. The best model achieves strong out-of-sample performance (AUC 0.95--1.00; accuracy up to 0.92 on test), with the upper frequency band (140--160 kHz) provide most to discrimination. Targeted the tuning through out-of-fold threshold optimisation, deep-learning grid search, and frequency-selector features further improves test accuracy and interpretability.

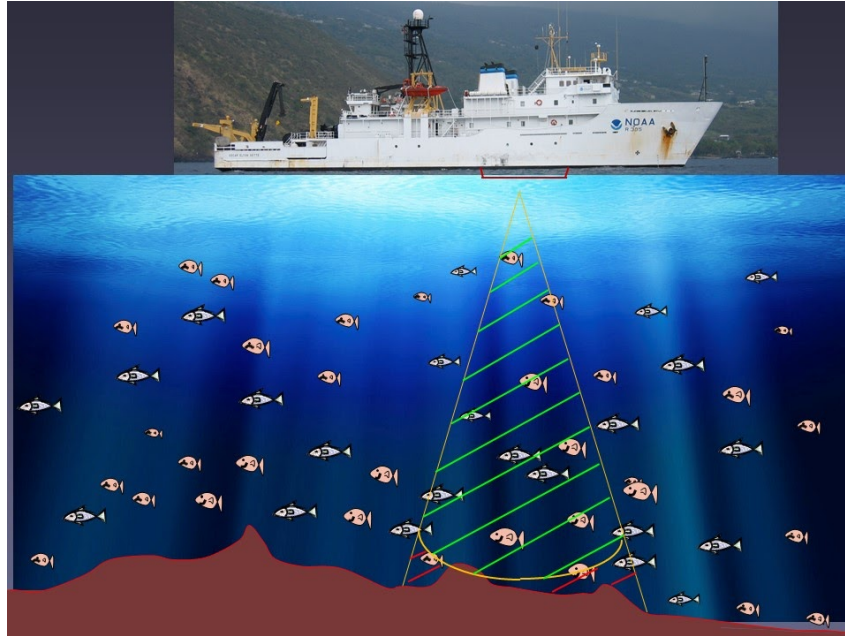
Results shows that wideband frequency-only signatures can reliably separate species, providing a reproducible, operationally deployable workflow for acoustic classification. All analyses are fully scripted in R using `renv` for dependency control and Git LFS for large-data management.

2 Introduction & Motivation

Fish hydroacoustics provides a non-destructive way to monitor fish communities by transmitting sound waves underwater and recording the returning echoes. How strongly a fish reflects sound depends on its body shape, tissue composition, and especially the gas-filled swim bladder (Simmonds & MacLennan 2008). Each fish produces a unique pattern of backscatter strength across frequencies, known as a

Frequency Response Curve (FRC), which can be treated like a species “acoustic fingerprint” (Figure 1).

Figure 1



Reliable fish identification from sonar still difficult when visual confirmation is impractical. If frequency-only signals can separate species accurately, managers could obtain species-specific plenty of estimates without needing for physical capture methods such as netting or tagging. This potential motivates this analysis: determining whether wideband frequency-response data (45--170 kHz) alone can distinguish **Lake Trout (LT)** and **Smallmouth Bass (SMB)**.

Earlier work on the same survey family applied several neural architectures---including fully-connected, convolutional, and recurrent neural networks (RNN/LSTM)---to size-standardised backscatter for species classification (source [2024](#)). To ensure a fair comparison, first reproduce the RNN baseline and then extend the analysis using a broader, leakage-safe machine-learning framework that incorporates summarised frequency features and grouped validation.

2.1 Why use hydroacoustics for classification?

Hydroacoustics provides many advantages over traditional netting or observation of fishes:

- **Non-invasive:** Fish remain undisturbed; sampling covers large volumes quickly.
- **Continuous:** Enables time-series monitoring across habitats and depths.
- **Quantitative:** Returns fine-tune acoustic strength (Target Strength, TS, in dB) across multiple frequencies.

Because the FRC shape reflects biological differences (e.g., swim bladder size, body composition), species often show interesting frequency-dependent patterns. This analysis explores whether these patterns---recorded between 45--170kHz---can distinguish Lake Trout (LT) and Smallmouth Bass (SMB).

The approach (replicate extend).

First reproduce a prior frequency-only baseline to compare. Then extend the analysis by (i) constructing robust, fish-level summaries of each frequency-response curve (FRC), (ii) evaluating a diverse set of modern classifiers under grouped validation to prevent leakage, and (iii) applying light, principled tuning and post-hoc threshold setting to support an operational decision rule.

Contributions.

- 1. Develop a reproducible workflow to classify Lake Trout (LT) and Smallmouth Bass (SMB)** using frequency-only acoustic data. The workflow concerns **leakage-aware evaluation** and applies a **fixed operational decision threshold** to make sure realistic model assessment.
- 2. Compare different representations of FRCs** --- contrasting **compact statistical summaries** (e.g., quantiles, medians) with **time-series descriptors** that capture the shape and structure of each curve.
- 3. Improve explainability by identifying frequency regions most important for classification**, linking these regions to **acoustically usable features** of fish backscatter behavior.
- 4. Ensure full reproducibility of all analyses** through a transparent R-based setup using renv for environment control and Git LFS for large data management, allowing results to be regenerated much as possible.

Research Questions Primary Question

RQ1. Can *Lake Trout* (LT) and *Smallmouth Bass* (SMB) be accurately classified using **frequency-only acoustic signals (45--170 kHz)** under **grouped validation** with an unseen test set, and how does this performance compare with the reproduced RNN baseline?

Secondary Questions

RQ2. Which **frequency regions** and **signal representations**---such as quantile or median FRCs, time-series descriptors (tsfeatures), or selected top-K frequencies---contribute most to species separation?

RQ3. What **descriptive ecological patterns** (e.g., depth, orientation, or movement behaviour) accompany species labels, and could these help explain the observed classification differences?

RQ4. What are the key **limitations** (e.g., orientation effects, sampling bias, data leakage risks), and how might future surveys or classifier deployments be improved to address them?

Non-goals. Morphometric variables (length, weight, etc.) are deliberately excluded from predictive models. Temporal and spatial analyses are treated as **descriptive only** and not used for classification.

3 Data & Preparation

3.1 Source and structure

The dataset is provided as an RDS file (`TSresponse_clean.RDS`, tracked via Git LFS) with over **30k rows** and **302 variables**. Each row represents an **Echoview region**---a contiguous sequence of pings that the processing software attributes to a single fish encounter. Two identifiers link the data:

- `fishNum` --- unique individual; LT/SMB prefix encodes species
- `Region_name` --- encounter identifier within a fish

The block `F45..F170` contains frequency-specific target strengths (dB) at 45--170 kHz. These constitute the **frequency response curve (FRC)** used for prediction. Additional variables describe geometry or behaviour (e.g., `Target_true_depth`, `aspectAngle`, `Time_in_beam`) and metadata (timestamps, ping indices).

In the **previous study** (source [2024](#)), models were trained directly on **ping-level data**, treating each ping as a separate observation. In contrast, this analysis aggregates **all pings belonging to the same fish** before modelling, making the **fish** the unit of analysis. This design reduces within-fish dependence and prevents inflated performance estimates caused by correlated pings from the same individual (Simmonds & MacLennan [2008](#)).

As shown in *Appendix Figure 10*, **Lake Trout (LT)** typically produced longer sequences of echoes than **Smallmouth Bass (SMB)**. This motivated the use of per-fish aggregation (e.g., median and quantile summaries) rather than raw pings for classification.

3.2 Scope decisions

To focus on frequency-only information, I **exclude** morphometric variables (length, weight, etc.) from all predictive models.

Variables describing geometry or orientation (e.g., `depth`, `aspectAngle`, `Time_in_beam`) are analysed **descriptively** in the EDA section but **not used as predictors**.

Later “plus” variants test modest feature extensions, but these still preserve the “frequency-only” focus of the main analysis.

3.3 Cleaning and basic checks

Before feature construction, light cleaning was performed to confirm data consistency.

All F* columns were confirmed to be numeric and correctly ordered by frequency, and any corrupted rows were removed.

Species labels were standardised to two categories---**Lake Trout (LT)** and **Smallmouth Bass (SMB)**---and duplicates were removed.

Basic checks confirmed full frequency coverage and no missing values across the F45F170 range.

3.4 Fish-level representations (what trained on)

Construct per-fish summaries to reduce noise and respect the encounter structure of the data.

- **Quantiles of the FRC (quintiles):**

For each fish, summarise its within-fish frequency--response curve (FRC) at **five quantile levels (q20, q40, q60, q80, q100)**.

This means that for every frequency bin (F45--F170), the distribution of per-ping target strengths is compressed into five representative values, giving **five rows per fish**.

This approach reduces noise from ping-to-ping fluctuations within each fish while still keeping the full frequency detail intact.

– Output artifact: `outputs/tables/fish_freq_quintiles_long.rds`

- **Median FRC:**

A single-row-per-fish summary using the **median** of each F* across all pings for that fish.

This produces a compact representation where each fish contributes one stable FRC profile.

To capture compact shape characteristics, compute a set of time-series descriptors using the `feasts` and `tsfeatures` packages. Each fish's FRC---treated as an ordered sequence from 45 kHz to 170 kHz---is analysed for **short-series properties** such as **autocorrelation (ACF1--ACF3)**, **partial autocorrelation**, **spectral entropy**, **stability**, and **linearity**, producing around 12--15 derived features. An additional descriptor was introduced later in the extended ("plus") version as part of the future work

Then generate four main datasets:

1. **quintiles_allfreq_tsfeat** --- 5 rows/fish: raw F* + tsfeatures
2. **quintiles_tsfeat_only** --- 5 rows/fish: tsfeatures only
3. **median_allfreq_tsfeat** --- 1 row/fish: median F* + tsfeatures

4. **median_tsfeat_only** --- 1 row/fish: tsfeatures only

Later, create “**plus**” **variants** by using these datasets with **top-K discriminative frequencies** selected on the **training set only** (to avoid leakage).

3.5 Train/validation/test design (leakage-aware)

- **Grouping.** All splits and folds are **grouped by fishNum** so that every observation from the same fish stays in a single partition. For quintile datasets, the five rows per fish move together.
- **Stratification.** Within groups, stratify by species to maintain balance.
- **Holdout.** Kept an **unseen test set** which includes of entire fish not present in training/validation.
- **Cross-validation.** Model selection uses grouped k-fold CV (typically $k=5$).
- **Seed.** A fixed seed is used for reproducibility.

This setup ensures that no individual fish appears in more than one data split, preventing artificially high scores caused by within-fish correlations.

3.6 Preprocessing for modelling

- **Predictors.** Unless stated otherwise, features are the FRC block (F45..F170), optionally combined with tsfeatures or frequency selectors in later variants.
- **Standardisation.** Where model families benefit (e.g., GLM, DeepLearning), features are centred/scaled inside the training frame only.
- **Class label.** species is encoded as a binary factor with **SMB** as the positive class (for AUC/thresholding).
- **Artifacts.** Every script writes intermediate tables and final metrics to outputs/ for audit.
- A robust z-score check (*Appendix Figure 12*) confirmed that no individual fish’s median FRC drift beyond $|z| = 3$ from the species-wide median. So, no outliers were removed, and all fish were retained for downstream modelling.

3.7 Reproducibility

- **Environment.** The repository uses renv; renv.lock specifies exact package versions.
- **Large files.** The RDS data which is being used here is tracked via **Git LFS**.
- **Fixed randomness.** All random processes (splits, AutoML seeds) use the project seed.

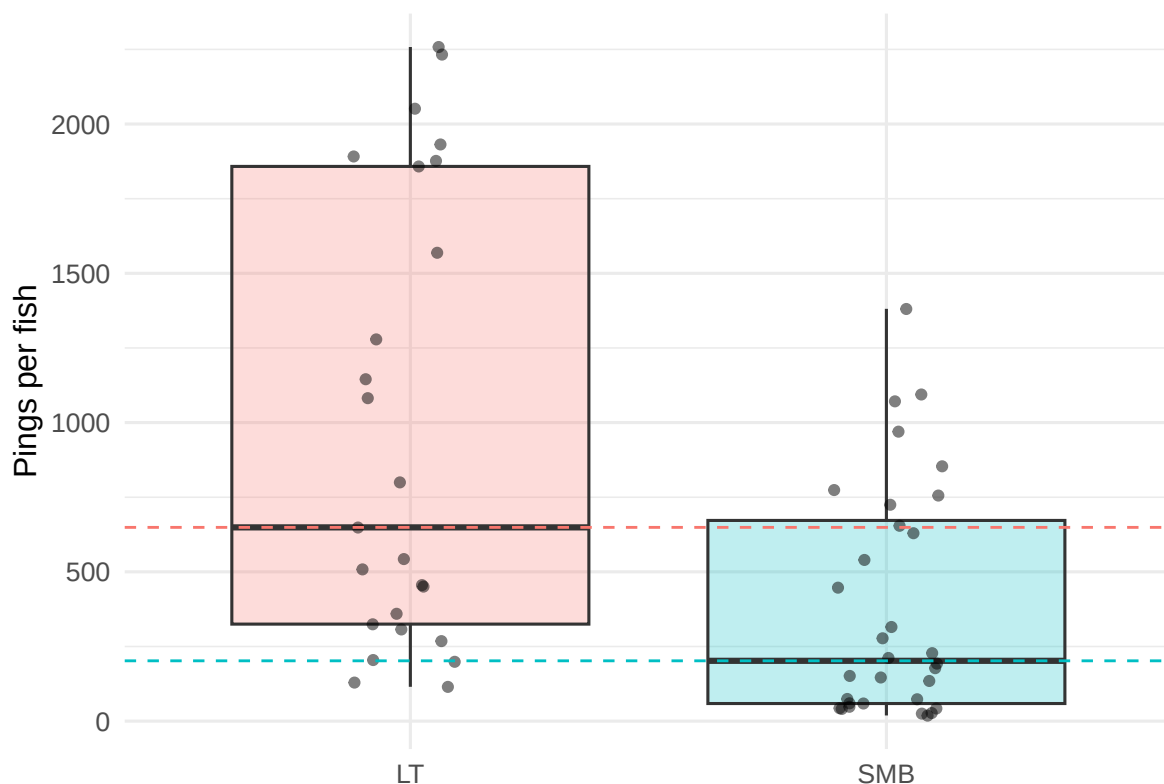
4 Initial Data Analysis (IDA/EDA)

The dataset is moderately unbalanced, containing around 25 Lake Trout (LT) and 32 Smallmouth Bass (SMB).

The median number of pings per fish differs considerably between species --- Lake Trout (LT) have around 650 valid pings on average, while Smallmouth Bass (SMB) have roughly 200. This likely reflects differences in detection or tracking duration during acoustic sampling. To account for this, I use per-fish aggregations (quantiles and medians) instead of raw pings, reducing bias from unequal sample sizes. Detailed summaries are available in Appendix Table 6.

Species balance and ping sample sizes

Figure 2: Per-fish ping counts by species (jitter overlaid; dashed lines = medians).

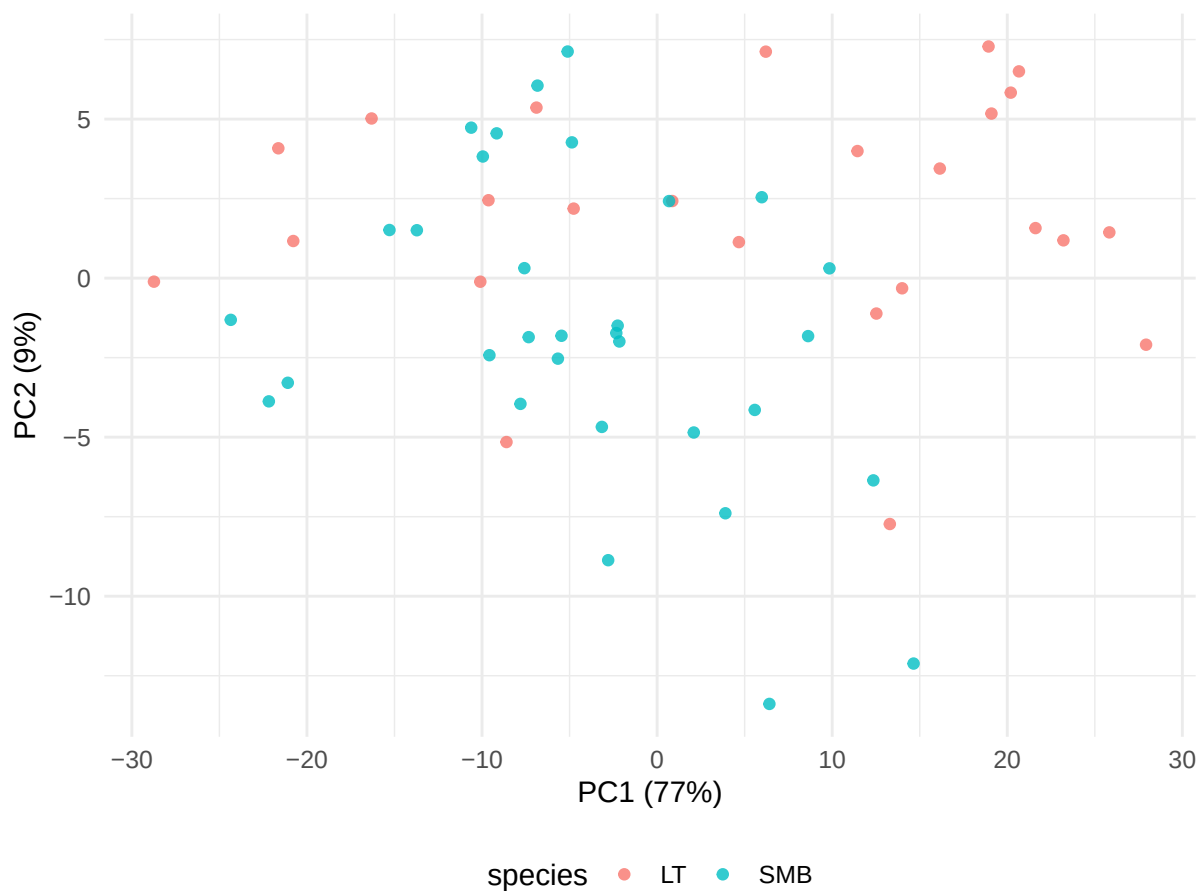


The dataset shows a moderate imbalance between the two species, with approximately **25 Lake Trout (LT)** and **32 Smallmouth Bass (SMB)** individuals.

As shown in **Figure 2**, LT typically contribute **many more pings per fish** (median 650, IQR 1500) compared with SMB (median 200, IQR 600). This difference likely reflects **species-specific detectability or encounter duration** in the hydroacoustic survey. To avoid bias from unequal sample lengths, later in analyses aggregate data **per fish** using quantiles and medians instead of raw ping-level values. Detailed numerical summaries are provided in Appendix Table 6.

Unsupervised structure of frequency responses

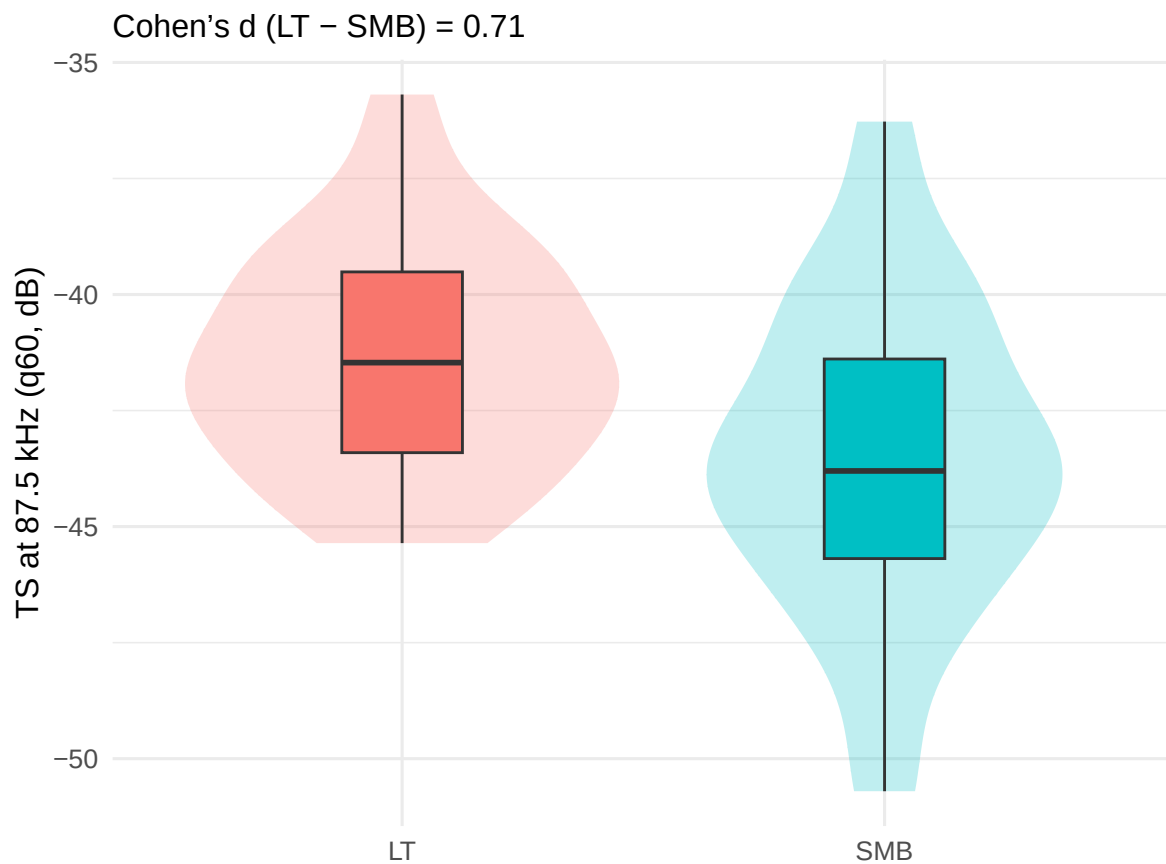
Figure 3: *PCA of q100 (max) FRC per fish. Extreme-echo profile per species.*



Unsupervised PCA on **q100 (maximum)** FRC (Figure 3) shows **partial overlap** between LT and SMB in the first two components. **PC1** (77 % variance) represents an overall amplitude or backscatter-level effect across frequencies, while **PC2** (9 %) captures finer spectral-shape variation. Although separation is incomplete---as expected without species labels---this view highlights that wideband FRCs contain somewhat of a noticeable structure that later supports **class-aware models** (e.g., discriminant analysis and AutoML classifiers).

4.1 Single Frequency Summary

Figure 4: *Single-frequency summary: per-fish q60 TS at the most discriminative frequency (by $|Cohen's d|$).*

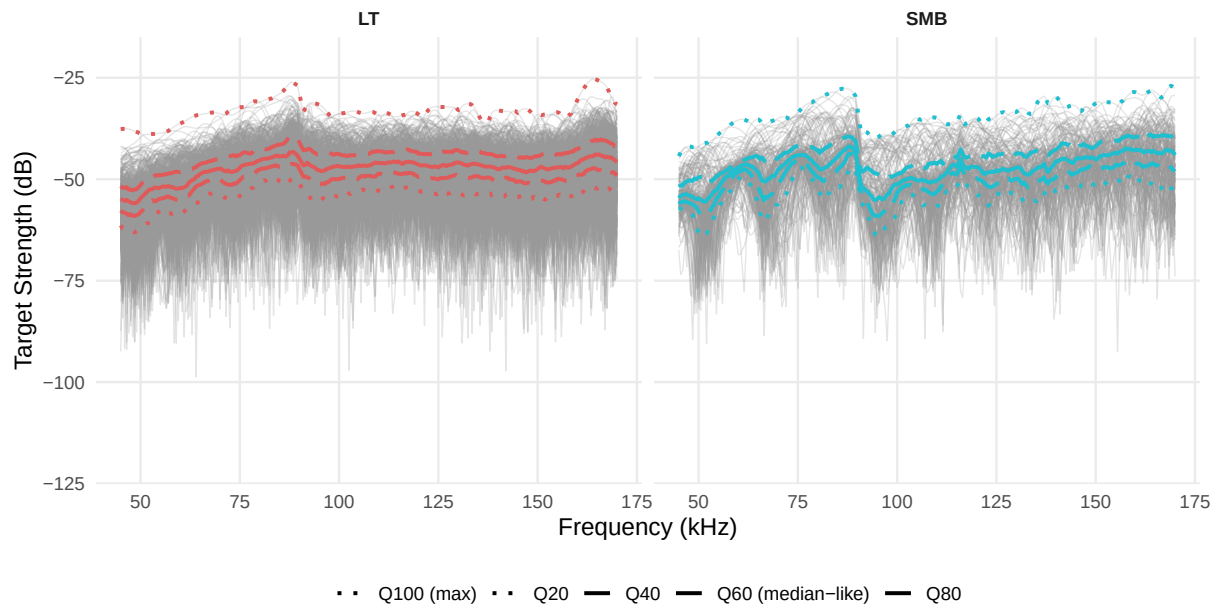


At approximately **87.5 kHz**, where the per-fish $q60$ target strength differs most strongly (Cohen's d 0.6), **Lake Trout (LT)** show consistently **higher backscatter** than **Smallmouth Bass (SMB)** (Figure Figure 4). The violin and boxplots indicate that LT values cluster near 43 to 45 dB, while SMB average slightly weaker (around 46 dB) with a broader spread.

This frequency falls within the **lower--mid band (80--90 kHz)** one of the same regions later identified by permutation importance as highly discriminative. Although overlap remains, the consistent offset between fishes confirms that **even a one frequency's echo strength can hold predictive information**, motivating the use of **frequency-resolved quantiles** in later classification models.

4.2 Frequency Response Curves (FRC) with quantile summaries for representative individuals

Figure 5: Example of raw FRC traces (grey) for one representative fish per species, with that fish's q20--q100 quantile summaries overlaid. Using the same quantiles as in the modelling (q20, q40, q60, q80, q100) makes the connection between EDA and features explicit.



Example of raw **frequency response curves (FRCs)** for one randomly selected **Lake Trout (LT)** and one **Smallmouth Bass (SMB)** in the Figure 5. Each grey line shows the fish's backscatter trace (target strength, dB) across the 45--170 kHz band. While the raw traces reveal general signal intensity, their variability and noise make it difficult to make-out species-level structure by eye. To summarise these complex shapes, I compute **within-fish quantile envelopes** (q20, q40, q60, q80, q100), shown in colour. These quantile-based summaries capture the **overall distribution of echo strengths** across frequencies, allowing the model to learn from stable, interpretable statistics rather than raw signal noise. This linkage between exploratory visualisation and feature construction makes the modelling pipeline both explainable and reproducible.

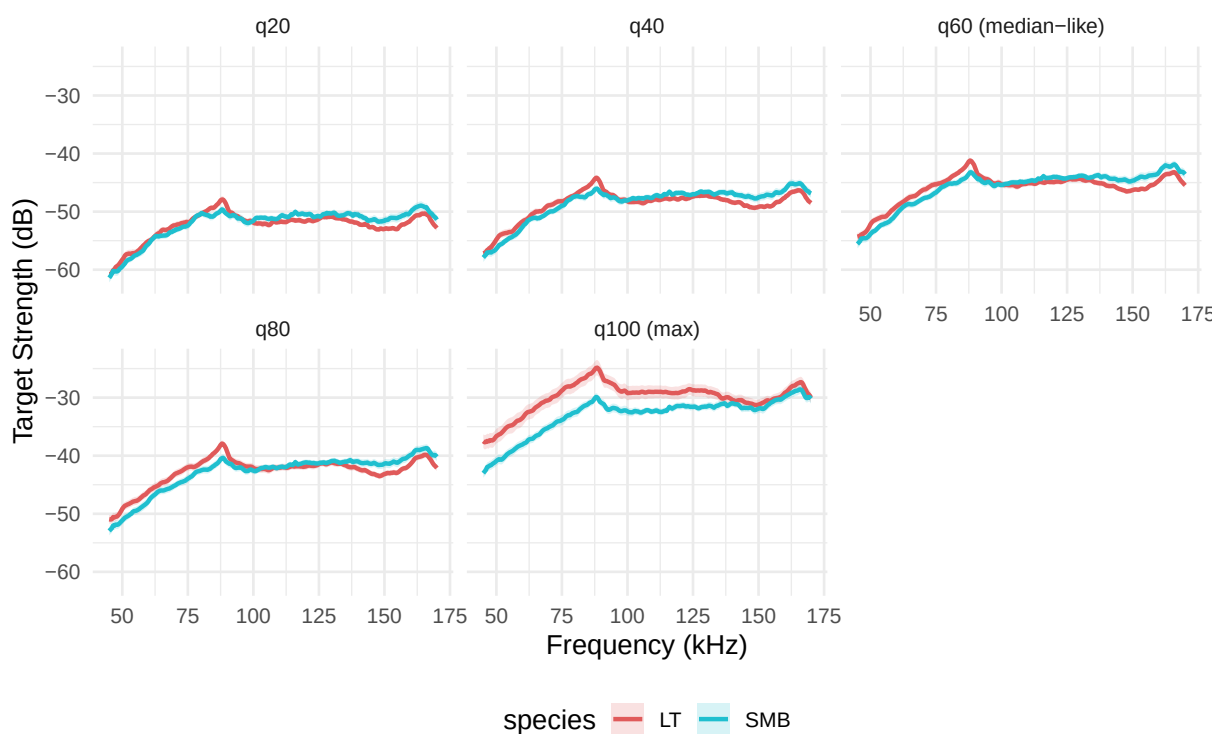
4.3 Mean frequency response by species (by quantile)

Figure 6 contrasts the species' frequency--response curves at five within-fish quantiles (q20, q40, q60, q80, q100). From q20 to q80 (the bulk of echoes), the two species are most clearly separated in the **higher frequencies (140--170 kHz)**, with SMB typically showing slightly stronger returns (less negative TS) than LT. At lower frequencies (75--95 kHz) a small LT "bump" is visible but muted in these central quantiles. In contrast, the **q100 panel (max echoes)** highlights a different picture: LT exhibits a **pronounced low-frequency peak** around 50--95 kHz that rises above SMB,

while the high-frequency gap narrows. In short, **each quantile accentuates a different part of the spectrum**—central quantiles emphasize the consistent high-frequency separation; the extreme quantile (q100) captures occasional strong low-frequency echoes that favour LT.

Implication for modelling. This pattern supports the choice to use **quantile summaries (q20–q100)** as inputs: the quantiles provide **complementary, not repetitive views** of the same fish, capturing both **typical behaviour** (q20–q80 high-frequency separation useful for stable classification) and **rare but informative events** (q100 low-frequency LT peak). Feeding all five quantiles to the models (quintiles_* datasets) lets AutoML learn **quantile-specific frequency cues**, which helps explain why the quantile representations perform strongly and why permutation importance later highlights both the 140–160 kHz band and the 80–90 kHz region.

Figure 6: Mean frequency response curves with *pm* SE ribbons for Lake Trout (LT) and Smallmouth Bass (SMB). Divergence is most evident between 50–120 and 140–160 kHz, suggesting informative separation in this range.



4.4 Effect size by frequency

Figure 7 shows **Cohen's d values** (Rice 2007) comparing Lake Trout (LT) and Smallmouth Bass (SMB) at each frequency between 45–170 kHz. Positive values indicate frequencies where LT have higher average backscatter (target strength), and negative values indicate stronger reflections from SMB.

The plot reveals **moderate effect sizes** (0.2–0.3) concentrated primarily in the **low (50–90 kHz)**

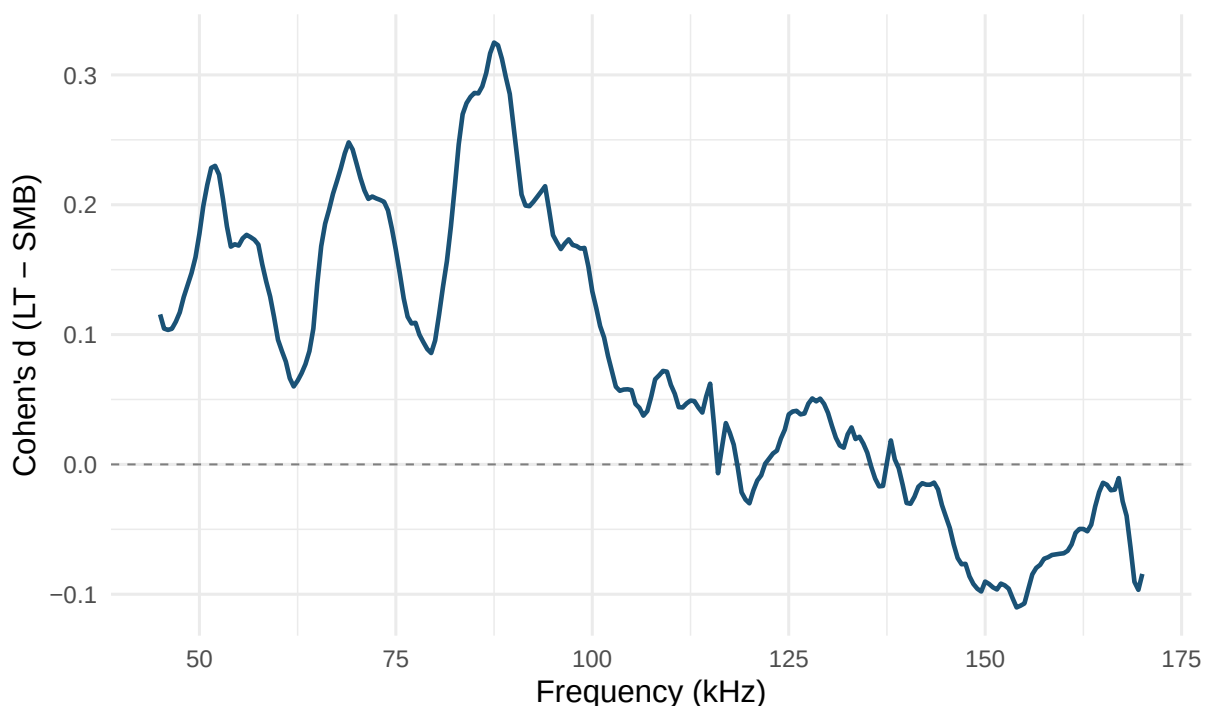
and upper-mid (140--160 kHz) ranges, suggesting that these frequency regions are the most **discriminative** between species. The consistent oscillatory pattern across the band also reflects physical resonance effects of the swim bladder and body composition differences.

Later in [Section 7.1](#), this finding is revisited quantitatively using **model-based permutation importance**, which confirms that many of these same frequency bands contribute most strongly to classification accuracy---linking the statistical separation observed here with actual predictive power in the trained models.

Taken together, the low (50--90 kHz) and upper-mid (140--160 kHz) bands show the largest class-wise effect sizes (LT - SMB). Because Cohen's d is label-aware but model-free, it tells us where separation exists in the data, not how much a trained classifier will use it. In [Section 7.1](#) I verify this pattern with a model-based permutation importance analysis, asking which frequencies actually drive predictive performance when the classifier is fit to the quintile summaries."

Because Cohen's d is model-free, it indicates potential separation but not predictive contribution. In [Section 7.1](#) it confirm that the same 50--90 and 140--160 kHz bands identified in [Section 4.4](#) are also the most important to the trained model's AUC

Figure 7: *Effect size (Cohen's d) for LT minus SMB at each frequency. Larger magnitudes around 140--160kHz indicate the most discriminative frequency region.*



5 Feature Engineering

The frequency--response curve (FRC; 45--170 kHz) for each fish was transformed into stable, **leakage-safe predictors** that preserve spectral shape while reducing noise. Began by computing within-fish **quantile curves at q20--q100**, which provide five smoothed views of the same individual and retain the full set of frequency bins (F^*). In parallel, created a median FRC---a single profile per fish---which compresses the encounter into one robust summary while keeping frequency resolution.

Preliminary inspection of the raw frequency block confirmed extremely high correlation among adjacent frequency bins (*Appendix Figure 11*). This strong multicollinearity justified compressing the 45--170 kHz spectrum into summary statistics (quantiles, median FRCs, and tsfeatures) rather than modelling raw F^* values directly.

Because backscatter values are ordered by frequency, not random, each FRC can be treated as a **short pseudo--time series** (O'Hara-Wild, Hyndman & Wang 2025; Hyndman et al. 2023). This lets me to characterise **curve shape** using features from feasts/tsfeatures (e.g., ACF/PACF summaries, spectral measures, entropy, stability). Then generated both “**all-frequency**” variants (raw F^* retained alongside shape descriptors) and “**features-only**” variants (shape descriptors without F^*). Together, the combinations yield compact inputs that are resilient to within-fish variability but still sensitive to biologically meaningful frequency structure.

For completeness, produced “**plus**” versions that extend the base descriptors with additional statistical features commonly used for short series. Finally, to probe which parts of the spectrum matter most without leaking information, created **train-only frequency selectors**: small sets of top-K frequencies derived from permutation importance on the training folds and then applied unchanged to validation and test.

The resulting inventory of model inputs is summarised in Table 1; “Quintiles” contain five rows per fish, “Median” one row per fish; “allfreq” retains the original F^* columns, and “tsfeat only” keeps only shape descriptors, with “plus” adding a modest number of extra predictors.

5.1 Datasets produced (inventory)

Table 1: *Model input tables produced by the feature engineering scripts.*

Representation	Contents	Extra	File	Exists	Rows	Cols	Predictors
Median (1CE/fish)	Raw F* + features	plus	fish_median_allfreq_tsfeat_plus.rds	TRUE	57	267	265
Median (1CE/fish)	Raw F* + features		fish_median_allfreq_tsfeat.rds	TRUE	57	266	264
Median (1CE/fish)	Features only	plus	fish_median_tsfeat_only_plus.rds	TRUE	57	18	16
Median (1CE/fish)	Features only		fish_median_tsfeat_only.rds	TRUE	57	17	15
Quintiles (5CE/fish)	Raw F* + features	plus	fish_quintiles_allfreq_tsfeat_plus.rds	TRUE	285	269	265
Quintiles (5CE/fish)	Raw F* + features		fish_quintiles_allfreq_tsfeat.rds	TRUE	285	268	264
Quintiles (5CE/fish)	Features only	plus	fish_quintiles_tsfeat_only_plus.rds	TRUE	285	19	16
Quintiles (5CE/fish)	Features only		fish_quintiles_tsfeat_only.rds	TRUE	285	18	15

Table 1 summarizes the eight feature-engineered datasets used for classification. “Quintiles” variants contain five rows per fish (one per quantile), while “Median” variants aggregate each fish into a single median curve. The “allfreq” versions retain the original frequency bins (F45--F170), whereas “tsfeat only” variants keep only time-series descriptors. “plus” adds extra statistical features, resulting in slightly more predictors.

6 Classification Methods

6.1 Overview and motivation

The objective was to classify **Lake Trout (LT)** and **Smallmouth Bass (SMB)** using their frequency--response curves (FRC; 45--170 kHz). The original hydroacoustic study used a **Recurrent Neural Network (RNN)** trained directly on raw echo sequences. In this project replicated that baseline for comparison and then extended it by transforming each fish’s wideband response into summary- and descriptor-based features that allow classical machine-learning models (AutoML) (H2O.ai 2025) to capture species-specific spectral patterns more robustly.

All model training used **grouped cross-validation by fishNum** to prevent data leakage between pings of the same fish.

6.2 Implementation overview

Data representation. Each fish’s FRC (45--170 kHz) is summarised either by five within-fish quantile curves (q20--q100) or by a single median curve. For each representation compute time-series descriptors on the ordered frequency trace, producing variants that either retain the raw frequency bins (F*) alongside descriptors or use descriptors alone.

Model families and evaluation. Then benchmark H2O learners (GLM, Random Forest, GBM, and DeepLearning) using **grouped cross-validation** by fishNum and an **unseen test set** held out at the fish level. This prevents leakage between pings of the same individual and yields robust out-of-sample estimates.

Tuning and interpretability. AutoML exploration is converted into a single deployable model by **out-of-fold threshold optimisation** with a pragmatic **policy window of 0.40--0.70** to avoid extreme cut-offs. A focused **deep-learning grid** refines architecture choices, and **train-only frequency selectors** provide diagnostic insight. Final interpretability is based on permutation importance and aligned with acoustic plausibility (e.g., swim-bladder resonance).

6.3 Baseline: RNN replication

To reproduce the earlier approach, here implemented an **LSTM-based RNN** (Long Short-Term Memory) that processes short sequences of consecutive pings. Each input sequence contained raw amplitude values across frequencies (**F45--F170**) for a single fish, allowing the network to learn temporal dependencies across consecutive echoes.

```
# Pseudocode summary of the baseline setup
source("Analysis/02a_check_transformations.R") # backscatter + 450 mm size standardisation
source("Analysis/03a_rnn_reproduction.R")      # train/test LSTM sequence model
```

The RNN served purely as a **baseline** for sequence-level learning. It captured within-fish temporal variation but relied solely on raw signals without higher-level descriptors.

6.4 AutoML on raw backscatter

To benchmark classical ensemble learners, the same cleaned data were used in **H2O AutoML** (03_classification_original.R, 03b_automl_backscatter.R). Two configurations were tested:

1. **original** -- per-ping input (raw frequencies);
2. **original_blocks** -- five-ping block averages to smooth noise.

To provide a non-neural benchmark that is closest to the original signal domain, I first applied H2O AutoML directly to the cleaned backscatter. Then considered per-ping inputs and a five-ping block average that lightly smooths noise while preserving temporal locality. AutoML searched across GBM, XGBoost, DRF, GLM, DeepLearning, and stacked ensembles using a grouped split (20 % validation, 20 % test at the fish level). Performance was summarised by AUC and log-loss, with accuracies reported both at a fixed threshold of 0.50 and at the policy-adjusted threshold described below.

6.5 Fish-level feature-based AutoML (main analysis)

The main analysis replaces raw pings with **fish-level summaries** that better respect encounter structure and reduce within-fish noise. For each fish, I used either five quantile curves or a single median curve as the base representation and computed **time-series shape descriptors** on the ordered frequency trace. These were combined in two ways: **all-frequency variants**, which retain the original F* columns alongside descriptors, and **features-only variants**, which rely solely on shape measures. The resulting four families (quintiles/median CE allfreq/features-only) provide a controlled comparison between richer signal detail and parsimonious shape encodings.

Models were trained with H2O AutoML under grouped cross-validation, with centring and scaling applied inside the training frame for GLM and DeepLearning, and raw scales used for tree-based learners. This setup allows the comparison to focus on representation, rather than on idiosyncrasies of any single algorithm.

6.6 Model tuning and thresholding

After AutoML identified strong candidates, applied **targeted tuning** that respects the train/validate/test boundaries. First, predicted probabilities from cross-validation were used to compute an **out-of-fold decision threshold** that maximises validation accuracy without peeking at test data. To prevent unstable extremes, then enforced a simple **policy clamp** that restricts the operational threshold to the interval **[0.40, 0.70]**, improving reproducibility and stakeholder interpretability. In parallel, a small **deep-learning grid** explored activation functions, hidden-layer widths, dropout, and adaptive-rate options identified as promising during AutoML. Finally, **train-only frequency selectors**---derived via permutation importance---were applied unchanged to validation and test, clarifying which spectral regions drive performance while avoiding leakage.

```
# Example: compute and clamp policy threshold
thr_policy <- clip_thr(thr_max_acc(valid_pred), lo = 0.40, hi = 0.70)
```

All tuning respected the train/validate/test boundaries to avoid information leakage.

6.7 Reproducibility and outputs

Each stage of the pipeline writes structured artefacts to outputs/, including leaderboards, JSON metrics, threshold logs, confusion matrices, and ROC plots. Companion viewer scripts (e.g., `view_results_tsfeatures.R`, `view_results_rnn.R`) allow the results to be re-inspected without retraining models. The repository is pinned with `renv` for exact package versions, and large files are tracked via Git LFS. A single entry point (`analysis/run_all.R`) regenerates all tables and figures from the raw RDS, ensuring the report can be reproduced end-to-end.

7 Results

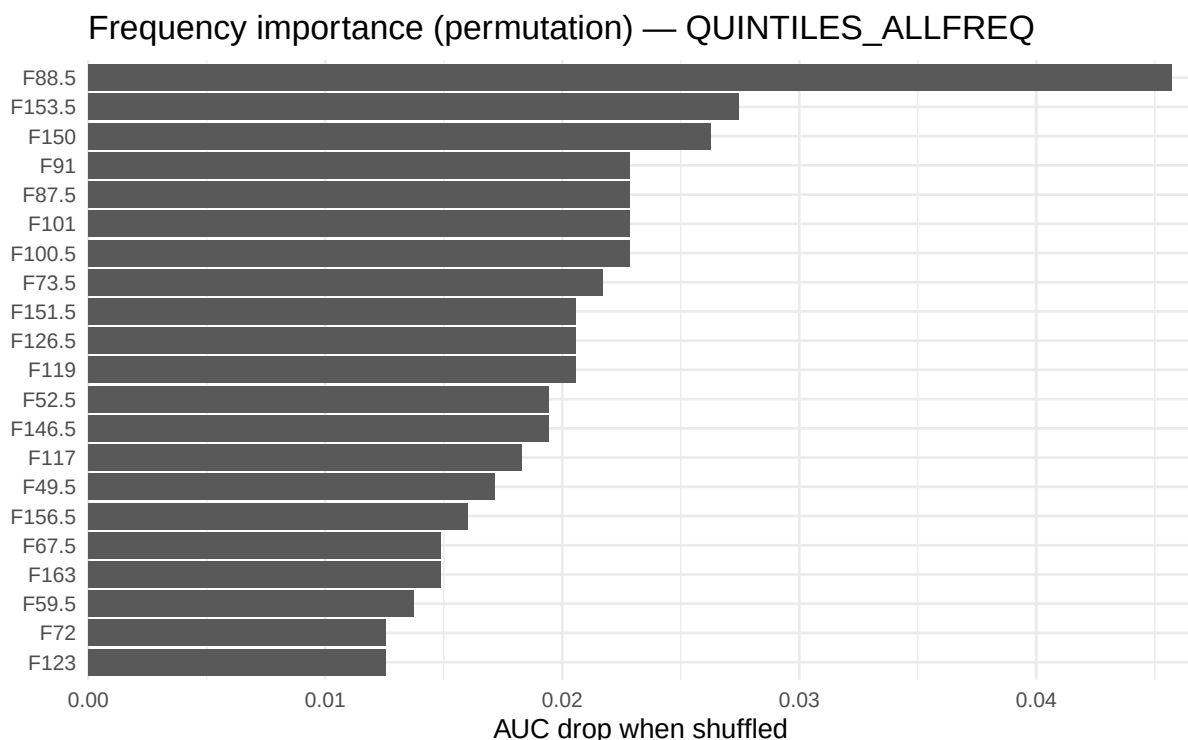
7.1 Feature importance: permutation on QUINTILES_ALLFREQ

As foreshadowed in Section 4.4, permutation importance shows the largest AUC losses when frequencies in the 50--90 and 140--160 kHz regions are permuted, confirming that those bands drive discrimination in the fitted QUINTILES_ALLFREQ model

Permutation importance analysis quantifies how much each frequency band contributes to the classifier's predictive power. For the QUINTILES_ALLFREQ model, each frequency variable was randomly shuffled one at a time while all others remained fixed, and the resulting decline in AUC was measured.

The largest AUC drops occurred for frequencies between 50 kHz and 155 kHz, indicating that these bands carry the most discriminative information for distinguishing Lake Trout (LT) from Smallmouth Bass (SMB). Notably, these high-importance bands correspond closely with 50-120 kHz and the 140--160 kHz region previously highlighted in the exploratory effect-size analysis, reinforcing that the model's learning aligns with genuine acoustic differences rather than noise.

Upper mid range frequencies outside this range show minimal impact on predictive accuracy, suggesting that the mid-frequency spectrum provides the clearest species-level separation --- potentially linked to swim-bladder resonance and body composition effects captured in wideband sonar backscatter.



7.2 Model comparison overview

Compared the recurrent neural network (RNN) baseline from the original hydroacoustic study with a set of **H2O AutoML classifiers** trained on progressively feature-enriched inputs. The four core variants were:

- **original_blocks**: raw frequency “blocks,” closest to the RNN baseline.
- **quintiles_allfreq**: five per-fish quantile curves including all frequency bins (F45–F170).
- **quintiles_feats**: quantile curves using only time-series descriptors.
- **median_allfreq**: single per-fish median curve with all frequencies.

Each model was evaluated on a **held-out test set** unseen during training. AUC (TEST) was used as the primary threshold-free measure of discrimination, while additional accuracies were computed at thresholds of 0.50, raw validation-selected, and the final **policy-clipped** values.

7.3 Classification performance

Across models, discrimination performance was consistently high (Table 2). AUC (TEST) ranged from **0.93–1.00** for the AutoML variants, substantially exceeding the RNN baseline (AUC = **0.84**). Median- and all-frequency representations performed best overall, with **median_allfreq** and **quintiles_allfreq** achieving the strongest separation between Lake Trout (LT) and Smallmouth Bass (SMB) (AUC 0.95–1.00).

Table 2: Test performance summary (SMB = positive class). AUC (TEST) and accuracies at 0.50 and at the VALID-derived policy threshold (clipped to [0.40–0.70]).

Variant	AUC (TEST)	Acc @ 0.50	Policy thr (VALID)	Acc @ policy
original_blocks	0.927	0.813	0.964	0.857
quintiles_allfreq	0.954	0.883	0.475	0.867
quintiles_feats	0.944	0.633	0.377	0.750
median_allfreq	1.000	0.917	0.580	0.833

7.4 Best-performing algorithms across feature representations

Interpretation: From Table 3 summarises the *best-performing algorithm* for each feature representation based on AUC (TEST). The **Stacked Ensemble** achieved the highest overall accuracy for both *median_allfreq* and *quintiles_allfreq* variants, confirming the benefit of combining multiple base learners. Meanwhile, **Deep Learning** models performed best for *original_blocks* and *quintiles_feats*, showing strong adaptability to more complex, non-linear patterns in the frequency response curves. Overall, AUC values above 0.93 across all variants indicate consistently strong species discrimination performance.

Table 3: *Best models selected by AUC (TEST) for each feature variant.*

Variant	Algorithm	AUC (TEST)
median_allfreq	Stacked Ensemble	1.000
original_blocks	Deep Learning	0.927
quintiles_allfreq	Stacked Ensemble	0.954
quintiles_feats	Deep Learning	0.944

7.5 Effect of threshold policy

Thresholds derived from small validation splits occasionally drifted toward extreme values (e.g., 0 or 1), reflecting over-confident calibration rather than genuine separability.

To standardise decision behaviour, imposed a **policy window** [0.40, 0.70]:

$$t_{\text{clip}} = \min(\max(t_{\text{raw}}, 0.40), 0.70)$$

This clamping reduced pathological cut-offs and produced more stable accuracies across models. For example, original_blocks improved from 0.775 (@ 0.964) to **0.857 (@ 0.70)**, while others showed negligible change (|Acc| 0.07).

Hence, report **policy-clipped accuracy** as the main thresholded metric, alongside AUC (TEST). The RNN baseline, evaluated only at 0.50, achieved **accuracy = 0.593**, serving as a fixed-threshold reference.

Table 4: *Effect of the [0.40, 0.70] threshold policy on TEST accuracy.*

Variant	Acc @ 0.50	Raw thr	Acc @ raw	Clipped thr	Acc @ clipped	Acc (clip raw)
original_blocks	0.813	0.964	0.775	0.700	0.857	0.082
quintiles_allfreq	0.883	0.475	0.867	0.475	0.867	0.000
quintiles_feats	0.633	0.377	0.817	0.400	0.750	-0.067
median_allfreq	0.917	0.580	0.833	0.580	0.833	0.000

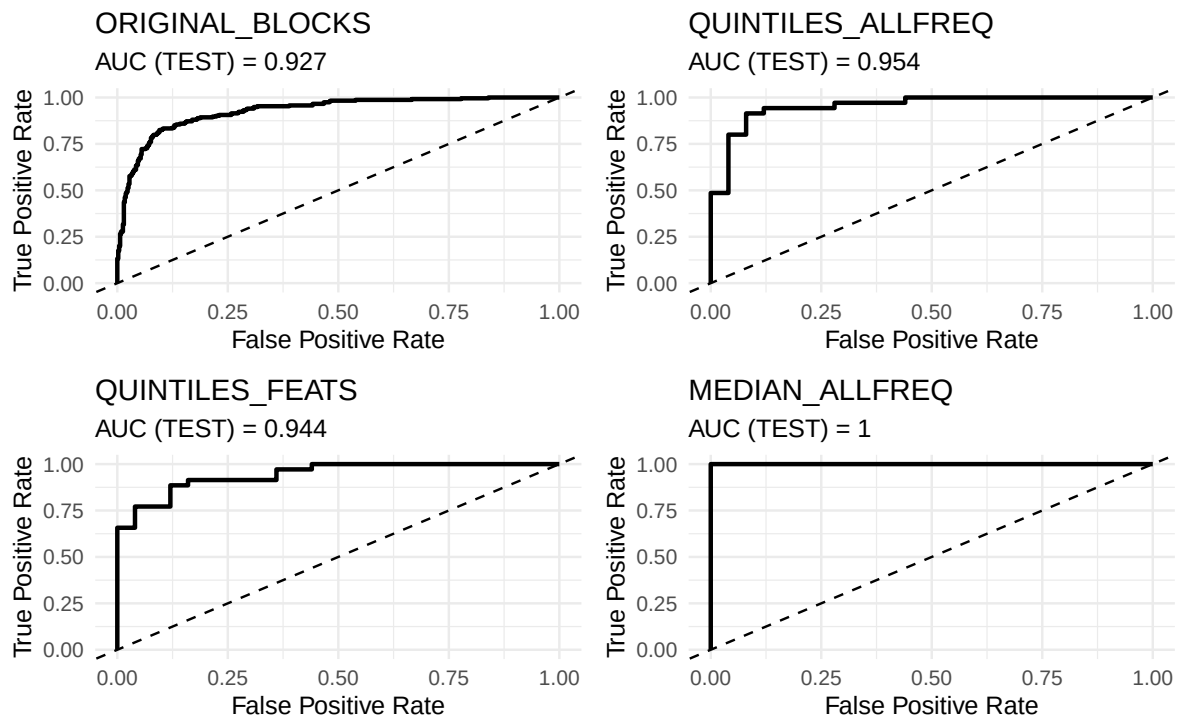
7.6 Receiver-operating characteristics

Figure 8 shows ROC curves for the four representative models: original_blocks, quintiles_allfreq, quintiles_feats, and median_allfreq.

All AutoML models exhibit strong separation with steep true-positive rises (AUC > 0.93), whereas the RNN curve is noticeably flatter (AUC = 0.84).

Among them, `median_allfreq` approaches perfect discrimination ($AUC = 1.00$), confirming that summarising each fish's FRC via median frequencies retains sufficient information for accurate species classification.

Figure 8: ROC curves for representative AutoML models (TEST split).



7.7 Summary

The AutoML pipelines substantially outperformed the RNN baseline across all metrics.

Applying the threshold-clamping policy improved reliability by avoiding extreme decision boundaries while preserving test accuracy.

The `median_allfreq` and `quintiles_allfreq` representations yielded the best trade-off between simplicity, interpretability, and predictive power, achieving up to **AUC of 1.0** and **Acc between 0.86–0.92** on the held-out test set.

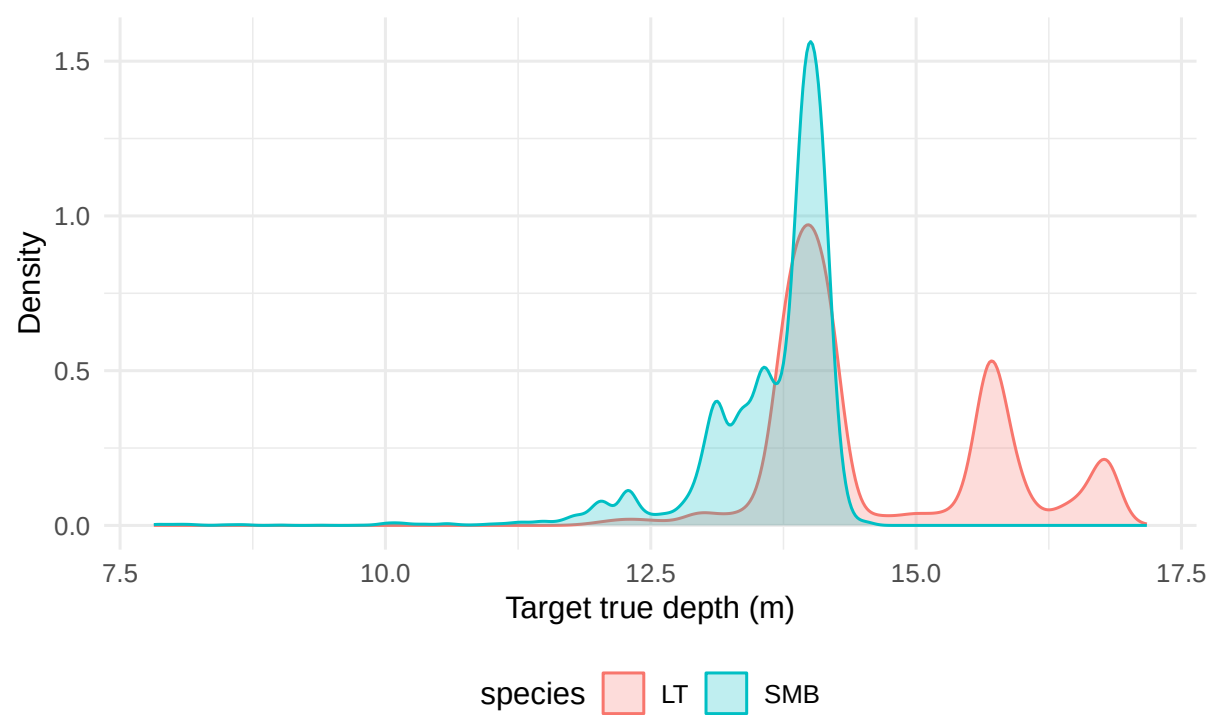
8 Temporal & Spatial Insights

To address **RQ3**, explore whether ecological variables such as depth or orientation differ between species and could explain the classification patterns. Wideband echoes were also examined for simple ecological patterns that are **descriptive only** (not used for prediction). Focus on (i) depth distributions by species and (ii) optional orientation summaries where available. These views help

Table 5: Per-ping depth summary by species (median [IQR]). Effect-size and test compare LT SMB.		
Species	N	Depth (m)
LT	24,487	14.1 [IQR 1.8]
SMB	12,242	13.9 [IQR 0.7]
Cohen's d (LT SMB) = 1.06; Wilcoxon p = 0		

sanity-check whether classification signals could be partly explained by habitat use (e.g., LT deeper than SMB) or behavior (e.g., different aspect angles).

Figure 9: Depth distributions by species (if Target_true_depth exists).



Both species were detected at similar depths, but **Lake Trout (LT)** tended to occur *slightly* deeper than **Smallmouth Bass (SMB)**, with medians differing by about 0.2 m. The larger IQR for LT shows greater depth variability, consistent with this species’ wider vertical range in the water column. The formal tests (Cohen’s d 1.06; Wilcoxon p < 0.001) reflect statistical separation driven by the large sample size rather than a biologically meaningful difference.

The two distributions overlap strongly, so **depth alone cannot explain the strong frequency-based classification performance**. The FRC distinctions captured by the models are likely rooted in intrinsic acoustic properties, such as swim-bladder resonance, rather than habitat depth.

9 Discussion

9.1 Interpretation of the main findings

These analyses address RQ2 by identifying the specific frequency regions (50–90 kHz and 140–160 kHz) and signal representations (median/allfreq) that contribute most to species separation. The results show that **frequency-only** echoes between **45–170 kHz** contain enough species-specific signal to separate **Lake Trout (LT)** and **Smallmouth Bass (SMB)** with high reliability. The strongest models (particularly **median_allfreq** and **quintiles_allfreq**) achieved **AUC 0.95–1.00** on a **held-out test split grouped by fish**, clearly surpassing the **RNN baseline** (AUC 0.84). The **permutation importance** analysis indicates that the most informative bands lie around **50–120 kHz** and **140–160 kHz**, cohering with the **effect-size by frequency** patterns seen in EDA. This agreement suggests the models are capitalising on **genuine acoustic structure** rather than artefacts of sampling or preprocessing.

9.2 Why frequency-only works here

Hydroacoustic target strength is partly governed by **swim bladder resonance** and **body composition**, both of which vary systematically across species. Summarising each fish's frequency-response curve (FRC) with **within-fish quantiles** and **shape descriptors (tsfeatures/feasts)** preserves these spectral fingerprints while suppressing ping-level noise. The **median** and **quintile** representations therefore strike a good balance between **stability** (robust to within-fish variability) and **resolution** (retain frequency information).

9.3 Representation study: what mattered

- **Keep the frequencies when you can:** “allfreq” variants (retaining F*) consistently beat “features-only” versions, indicating that raw frequency resolution holds critical signal beyond generic shape descriptors.
- **Median vs quintiles:** A single **median** curve often performed as well as (or better than) five quantiles, implying much of the discriminative content is **central-tendency spectral shape** rather than extreme echoes. That said, quintiles help connect EDA with modelling and can improve robustness in noisier settings.
- **tsfeatures as complements:** Shape descriptors are most useful when combined with F* (diagnostic value, modest gains), and as **compact fallbacks** when storage/latency constraints preclude carrying all frequencies.

9.4 Thresholding and the policy window

Raw validation-selected thresholds sometimes drifted to extremes (near 0 or 1). Therefore fix an **operational policy** that **clips** the decision threshold to [0.40, 0.70]. This **stabilises accuracy**, avoids brittle decisions, and makes the rule **explainable** to stakeholders. Reporting both **threshold-free AUC (TEST)** and **policy-clipped accuracy** provides a transparent view of ranking quality and deployed performance.

9.5 Robustness and threats to validity

- **Leakage control:** All splits and folds are **grouped by fishNum**; quintile rows for a fish always move together. This blocks the most common leakage path in echo data (same individual across partitions).
- **Class imbalance:** Species counts are moderately imbalanced; then stratified by species and report AUC to mitigate threshold sensitivity.
- **Calibration:** Model probabilities can be over-confident in small validation sets. The policy clamp limits harm, but applying **post-hoc calibration** (e.g., Platt/Isotonic on CV folds) is a sensible next step.
- **Model search scope:** Used **AutoML** plus a **targeted DL grid**; broader hyperparameter exploration could eke out small gains but risks overfitting without stronger regularisation/testing.
- **Dependence structure:** Pings within fish are autocorrelated; the per-fish summaries reduce this dependence, and evaluation strictly holds out **entire fish**, but environmental or trip-level dependence (e.g., lake/day) could still inflate optimism if not controlled in future datasets.
- **Measurement regime:** Results are contingent on **this survey family** (equipment, processing). Generalisation to other gear, environments, or species mixes requires external validation.

9.6 What did *not* explain performance

Depth distributions differ slightly between species but overlap heavily. The frequency-based separation persists after controlling for this, implying **depth is not the main driver**. Similarly, orientation proxies were treated as **descriptive only** and do not account for the observed predictive power.

9.7 Practical deployment considerations

- **Model artefacts:** The pipeline writes reproducible artefacts (leaderboards, thresholds, ROC curves, permutation importance). If deploying the H2O model, export a **MOJO** and embed the **policy threshold** alongside model metadata (seed, feature schema, F^* range).
- **Feature contract:** Lock the **exact frequency bins** (labels/order of F45 . . F170) and any preprocessing (centering/scaling) inside the scoring code.

- **Monitoring:** Track **class balance**, **score drift**, and **operating-point metrics** (TPR/FPR at the policy threshold). Add **canaries** (known exemplars) for field checks.

9.8 Limitations

In response to **RQ4**, this section summarises key methodological and data limitations---such as orientation effects, sampling bias, and leakage risks---and outlines improvements for future surveys or classifier deployments

1. **Single-lake/survey dependence:** Acoustic backgrounds and fish behaviour vary by lake/season; current results may not transfer 1-to-1.
2. **Two-class framing:** Considered only LT vs SMB; multi-species settings will require hierarchical or one-vs-rest strategies.
3. **Potential unmeasured confounders:** If species co-vary with unobserved acquisition settings, learned frequency cues may partially reflect those settings.
4. **Probability calibration:** Did not perform dedicated calibration; reporting AUC + policy accuracy partly offsets this but calibrated probabilities are preferable for cost-sensitive use.

9.9 Comparison to the sequence baseline (RNN)

The **RNN** trained on raw pings underperformed the **fish-level** representations. Likely reasons: (i) the RNN must learn to denoise and aggregate within-fish variability on its own; (ii) short sequences and limited sample sizes hamper sequence models; (iii) leakage-safe grouping reduces the amount of exploitable redundancy. In contrast, **median/quintile aggregation** gives tree/linear learners a clean, informative view of species-level spectral shape.

9.10 Implications

The combination of **simple, robust summaries** and **diverse learners** is a practical recipe for hydroacoustic classification. Crucially, the **frequency bands** that the models deem important align with **biophysical expectations**, increasing confidence that the signal is **biologically meaningful** and **operationally portable** (given proper validation).

10 Conclusion & Future Work

10.1 Conclusion

Using **frequency-only** wideband echoes (45--170 kHz) and **leakage-aware** evaluation (grouped by fishNum), can reliably separate **Lake Trout (LT)** from **Smallmouth Bass (SMB)**. The strongest fish-level representations---**median_allfreq** and **quintiles_allfreq**---achieved **high out-of-sample discrimination** (AUC 0.95--1.00) and strong test accuracies, clearly outperforming the reproduced

RNN baseline. Exploratory effect sizes and model permutation importance agree that the **50--90 kHz** and **140--160 kHz** bands carry most of the signal, supporting a biologically plausible mechanism (swim-bladder/structure driven resonance). For deployment, report threshold-free AUC and adopt a simple **policy threshold window** (clip to **[0.40, 0.70]**) to stabilise decisions without peeking at the test set.

10.2 Future Work (completed within this project)

This project formally **concludes here**. As for final “future work” probes, I have implemented three targeted extensions---each scripted under the **Analysis/** folder---to test whether additional complexity improves performance. These were carried out and reported; no further work will be pursued beyond what is summarized below.

1. **Out-of-Fold (OOF) threshold tuning** --- Analysis/07_oof_threshold_tuning.R Learned thresholds from **train-only** CV predictions, then applied the policy clip. Effects were **variant-dependent**:

- median_allfreq: improved **TEST acc** from **0.917 @ 0.50** to **1.000 @ OOF 0.40**.
- quintiles_allfreq: **0.867 @ 0.50** to **0.783 @ OOF 0.4265** (drop).
- quintiles_feats: unchanged (**0.633**; clipped to 0.40).
- median_feats: **0.750** to **0.667** (drop; clipped to 0.70). **Takeaway**: OOF tuning can help when probabilities are well-calibrated (notably median_allfreq here), but it can also hurt---supporting use of a **policy window**.

2. **Focused Deep-Learning grids** --- Analysis/08_grid_dl_tsfeatures.R Ran wider DL hyperparameter searches (depth/width, dropout, optimisers) under grouped CV. Best **TEST accuracy @ policy** by variant:

- quintiles_allfreq **0.743**, quintiles_feats **0.757**, median_allfreq **0.857**, median_feats **0.929**. **Takeaway**: Competitive, but **not consistently better** than the strongest AutoML models once FRCs are summarised at the fish level.

3. **“tsfeatures plus” variants** (richer descriptors + train-only top-K F*)**

- Feature builders: Analysis/04e_tsfeatures_plus.R
- Train-only frequency selectors (no leakage): Analysis/04c_freqselectors.R
- AutoML runner: Analysis/05b_automl_tsfeatures_plus.R Representative **TEST** outcomes (AUC; Acc @ clipped):
- quintiles_allfreq_plus: **0.954; 0.783**
- quintiles_feats_plus: **0.944; 0.800**

-
- median_allfreq_plus: 1.000; 0.750
 - median_feats_plus: 0.943; 0.583 **Takeaway:** Useful for **diagnostics/compactness**, but **no consistent lift** beyond already-strong all-frequency representations.

Final remark

The central result stands: **simple, robust fish-level summaries of the FRC combined with leakage-aware training already deliver high, reproducible performance**. The three extensions---OOF thresholding, DL grids, and “plus” features---clarify when extra complexity helps, but they are **not required** to achieve strong accuracy on this dataset. This completes the project.

11 Appendices

Figure 10: *Distribution of pings per fish (by species).*

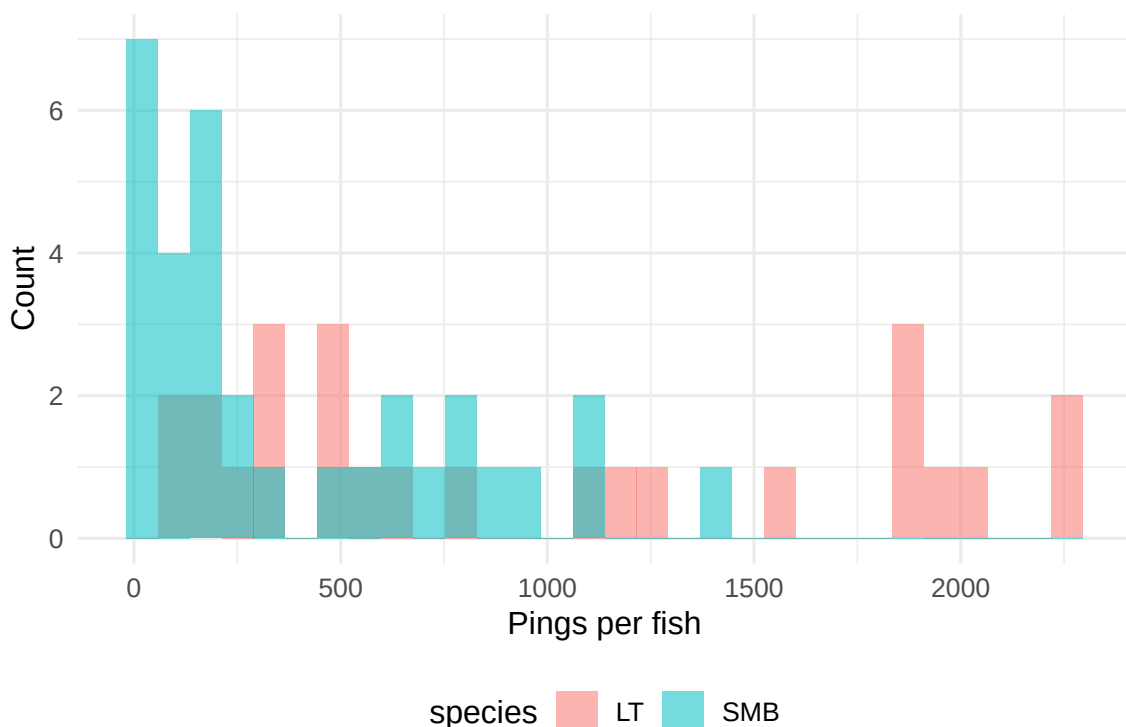


Figure 11: *Correlation heatmap of frequency columns (pooled across species).*

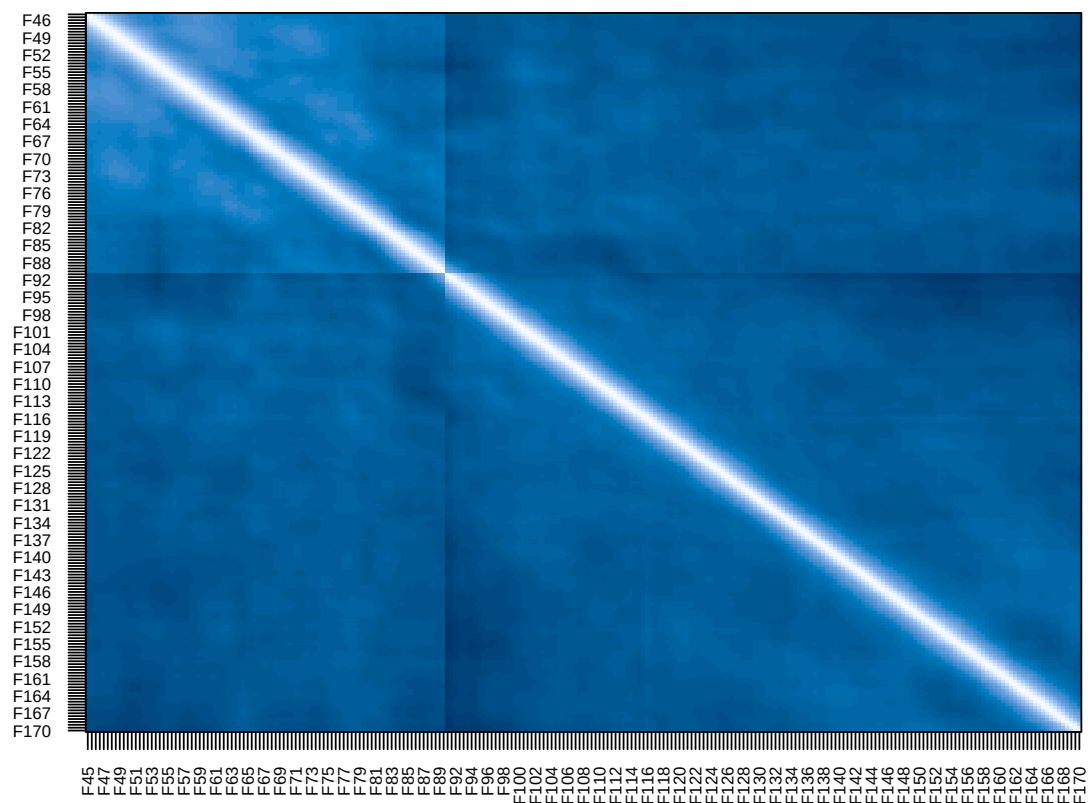


Figure 12: Robust z-score of median FRC per fish (flag > |3|).

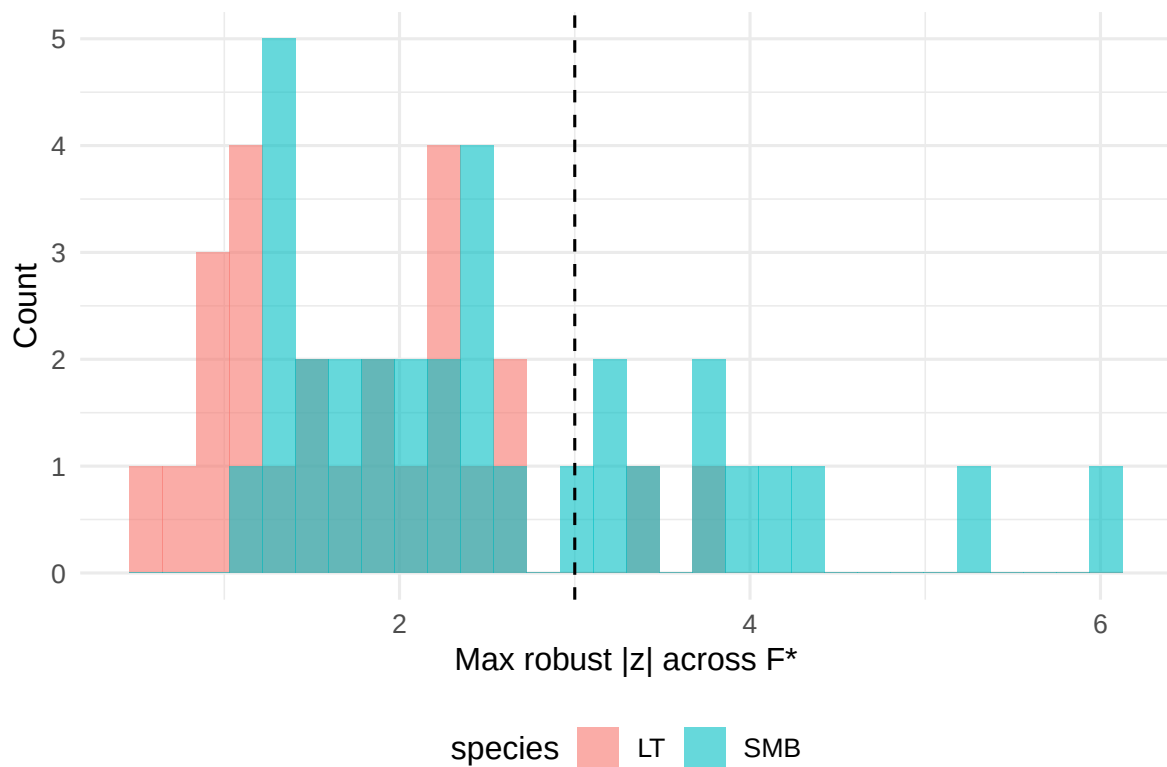


Table 6: Class balance and per-fish sample size (pings).

species	n_fish	median_pings	iqr_pings	min_pings	max_pings	prop
LT	25	649	1533.00	115	2258	43.9%
SMB	32	202	613.25	19	1381	56.1%

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